



# **Stingray™ PS225**

## **NVRAM Access Guide**

### **Application Note**

---

Broadcom, the pulse logo, Connecting everything, NetXtreme, Stingray, Avago Technologies, Avago, and the A logo are among the trademarks of Broadcom and/or its affiliates in the United States, certain other countries, and/or the EU.

Copyright © 2019-2020 Broadcom. All Rights Reserved.

The term “Broadcom” refers to Broadcom Inc. and/or its subsidiaries. For more information, please visit [www.broadcom.com](http://www.broadcom.com).

Broadcom reserves the right to make changes without further notice to any products or data herein to improve reliability, function, or design. Information furnished by Broadcom is believed to be accurate and reliable. However, Broadcom does not assume any liability arising out of the application or use of this information, nor the application or use of any product or circuit described herein, neither does it convey any license under its patent rights nor the rights of others.

---

# Table of Contents

1

About This Document

4

1.1

Purpose and Audience

4

2

Introduction

4

2.1

Supported NVRAM Devices

4

3

NVRAM Programming and Configuration

5

3.1

Firmware Update

5

3.2

Typical NVRAM Operations

5

3.3

NVRAM Configuration

7

3.3.1

NVRAM on BCM958802A804xE

7

3.3.2

NVRAM Configurable Item Descriptions

8

Appendix A: Acronyms and Abbreviations

14

Revision History

15

# 1 About This Document

## 1.1 Purpose and Audience

This document provides information necessary to program an external nonvolatile memory (NVRAM) and to access the various NVRAM configuration options available for the Broadcom® NetXtreme® BCM5880X using the Broadcom diagnostic utility called bnxtnvm.

## 2 Introduction

The Broadcom NetXtreme-S BCM5880X SmartNIC requires an external NVRAM component to store various device firmware and configuration options. This document describes the background design philosophy and changes over NVRAM designs from previous generations of controller products.

This document applies to all Broadcom NetXtreme S-series (BCM5880X) SNIC controller devices.

### 2.1 Supported NVRAM Devices

The NVRAM devices supported by the NetXtreme S-based platforms are listed in [Table 1](#).

**Table 1: Supported Serial Flash NVRAM**

| NVRAM Manufacturer | Part Number     | Device Size |
|--------------------|-----------------|-------------|
| Macronix           | Part MX25L6406E | 64 Mbit     |
| Micron             | N25Q064A        | 64 Mbit     |
|                    | N25Q0128A       | 128 Mbit    |
| Winbond            | W25Q64FV        | 64 Mbit     |
|                    | W25Q128FV       | 128 Mbit    |

## 3 NVRAM Programming and Configuration

### 3.1 Firmware Update

After a device's NVRAM has been programmed, and the contained firmware and configuration is operational, reliably update the version of the various firmware components without risk to the operational state of the device. A processor reset to allow a new firmware version to take effect (execute), must be graceful and only occur when there are no expected adverse effects on the system.

### 3.2 Typical NVRAM Operations

**NOTE:** `bnxtnvm` is an end-user tool that is used to perform operations such as: listing NVRAM directory contents, viewing details about each item in the NVRAM package file, removing items from the NVRAM, installing and/or upgrading firmware components, verifying validity/authenticity of the NVRAM items in the NVRAM package file, and comparing the contents of the package files to NVM items in the NVRAM. Refer to the README.txt included with this tool for additional details.

To be enumerated correctly in an OS environment, the NVRAM must be programmed with boot code and contain a valid set of default configuration.

**NOTE:** Some NVRAM operations may no longer be available due to secure firmware update implementation.

#### ■ `bnxtnvm`

Use the `bnxtnvm` tool on the BCM5880X to modify the `nvm` config of the MAIA and host-side PFs. The `bnxtnvm` tool provides various options other than `nvm` config changes. Use `bnxtnvm -help` to view the supported options.

```
root@bcm958802a8048c:~# bnxtnvm -h
basicusage: bnxtnvm <command> [-option [...]] [[filename] [...]]
```

Broadcom NetXtreme-C/E Non-volatile Memory Management Utility

commands:

|                        |                                                                           |
|------------------------|---------------------------------------------------------------------------|
| <code>install</code>   | install package file(s) into NVM of device(s)                             |
| <code>upgrade</code>   | install newer package file(s) into NVM of device(s)                       |
| <code>verify</code>    | verify contents (checksums, signatures, etc.)                             |
| <code>pkgver</code>    | display [installed] package version information                           |
| <code>devid</code>     | display [target] PCI device IDs                                           |
| <code>list</code>      | display directory entries and item version details                        |
| <code>view</code>      | display package/item details or contents                                  |
| <code>erase</code>     | erase an item from NVM                                                    |
| <code>backup</code>    | backup NVM contents into a package file                                   |
| <code>compare</code>   | compare NVM contents with a package file                                  |
| <code>listdev</code>   | discover and display supported controller devices                         |
| <code>status</code>    | display controller/NVM/firmware status                                    |
| <code>version</code>   | display program version details                                           |
| <code>getoption</code> | get current NVM configuration option settings for a device                |
|                        | syntax: <code>getoption=&lt;option name&gt;[:index]</code>                |
|                        | examples: <code>getoption=enable_sriov</code>                             |
|                        | <code>getoption=num_vf_per_pf:1</code>                                    |
| <code>setoption</code> | set NVM configuration option for a device                                 |
|                        | syntax: <code>setoption=&lt;option name&gt;[:index]#&lt;newval&gt;</code> |
|                        | examples: <code>setoption=enable_sriov#0</code>                           |
|                        | <code>setoption=num_vf_per_pf:1#8</code>                                  |

```

optionhelp           Detailed help for each NVM configuration option
saveoptions          Save options (config parameters) <fname> [#ports] [#fcns]
resmgmt              Resource Management to Get/Set Resources
get_backup_power_config  get backup power configuration of the device
set_backup_power_config  set backup power configuration of the device
coredump             Retrieves coredump data into a .core file
reset                reset the firmware
device_temperature    Displays device temperature in Celsius
cfgtunnel            Custom tunnel configuration (Query, free Port and Set Port)
add_tunnel_redirect   Add tunnel type redirection
del_tunnel_redirect   Remove tunnel type redirection
moduleinfo           displays the module information
ether_serdes_eye_diagram  Display ethernet serdes eye diagram
                        syntax: ether_serdes_eye_diagram [-margin|-v|-V]
pci_eye_diagram       Display pci eye diagram
                        syntax: pci_eye_diagram <lane number> [-margin|-v|-V]
reset_ap             reset application processor
pcie_counters        Displays pcie counters of an adapter
set_vf_queues        set VF queues parameters
get_vf_queues        get VF queues parameters
set_mtu              set MTU of requested PF
get_mtu              get MTU of requested PF
vf                  configures and queries the trusted VF for a given VF index
elog_dump            Reads error logging contents in NIC QSPI and writes into a elog_dump.bin file

```

#### command options:

```
-dev=<device>          specify controller device to act on (?=list)
```

#### console output options:

```

-v[v][v]             enable extra console output (increase verbosity)
-V                   enable maximum verbosity of console output
-h                   display help for basic command-line syntax
-H                   display help for advanced command-line syntax

```

**NOTE:** Some commands require root/administrator privileges.

```

root@bcm958802a8048c:~# bnxtnvm version
Broadcom NetXtreme-C/E Non-volatile Memory Management Utility
Version 214.1.59.40, (BMAPI 214.0.42.0) for Linux

```

#### 1. Prepare an NVRAM configuration file.

A sample or default configuration file can be generated from an existing NVRAM content or provided to you in your software kit, available from the customer support portal (CSP). The CFG file is a text file containing a comment section, followed by a shared configuration, configuration for device 1, and configuration for device 2. If NPAR is enabled, an additional function config block (device 3, 4 ...and so on) is added for each enabled NPAR PF.

**NOTE:** Lines prepended with a # are comments and meant to describe each corresponding item and its meaning. The 559 configurable items available are organized into multiple groupings for ease of access. Lines without a # are actual settings and take the following format:

```

{1=00:10:18:AA:BB:CC}\
Or
{item number = value}\

```

The entry above sets item 1, the MAC address, to value 00:10:18:AA:BB:CC. For a dual-ports device, the shared items apply to both ports. The items under device 1 apply to the first port, while items under device 2 apply to the second port.

## 2. Program the boot code:

```
$ bnxtnm -dev=enp81s0f0 install nitro.pkg  
Syntax : bnxtnm -dev=<device/interface name> install <package name>
```

This command launches engineering mode and programs a packaged firmware named nitro.pkg to NVRAM. At the time of this document's release, a packaged firmware typically includes the Falconcore firmware, boot code, and PCIe firmware. Similarly, other firmware components can be programmed using this command.

**NOTE:** nitro.pkg is the boot code for the BCM5880X silicon. Using the incorrect bootcode file can render the controller inoperable.

This command programs the data and executable object code to the NVRAM. The filename specifies the name of the file that contains the appropriate image.

## 3. Configure specific items:

```
$ bnxtnm -dev=enp81s0f0 setoption=num_vf_per_pf:1#8  
Syntax : bnxtnm -dev=<device/interface name> setoption=<option name>:<ID>
```

This sets configuration item num\_vf\_per\_pf, the number of VFs for the PF 1 to 8. The bnxtnm command with setoption=<item>:<id>#<value>, where id indicates device the (PF) ID.

## 4. Save the CFG options:

```
$bnxtnm -dev=enp81s0f0 saveoptions mycfg.txt  
Syntax : bnxtnm -dev=<device/interface name> saveoptions <file name>
```

This command dumps the current configuration settings to the text file mycfg.txt. This configuration file can be used to program another NVRAM with this configuration.

## 5. Save the NVRAM image to file:

```
$ bnxtnm -dev=enp81s0f0 backup  
Syntax : bnxtnm -dev=<device/interface name> backup
```

This command saves the current NVRAM content to a file called 000AF7F1F810.pkg where the name format of the file is <mac address of first PF>.pkg. The entire content of NVRAM is saved to 000AF7F1F810.pkg. This file can be used as a master image for duplication to another NVRAM part.

## 6. Program the saved NVRAM image:

```
$bnxtnm -dev=enp81s0f0 upgrade mynvm.pkg  
Syntax : bnxtnm -dev=<device_interface_name> upgrade <package_file>
```

This command programs the saved NVRAM image mynvm . pkg to NVRAM starting at offset 0. By default, this command first erase all directories in NVRAM and recreates items in NVRAM from the content of mynvm . pkg. As a result, the MAC address in current NVRAM is overwritten with the MAC address from mynvm . pkg.

# 3.3 NVRAM Configuration

## 3.3.1 NVRAM on BCM958802A804xE

Refer to the product layout design document for the location of the NVRAM flash device on the board.

### 3.3.2 NVRAM Configurable Item Descriptions

#### NOTE:

- [F] – Indicates the corresponding configurable item applies only to the specified function.
- [P] – Indicates the corresponding configurable item applies only to the specified network port.
- [S] – Indicates the corresponding configurable item is shared across the entire device.

#### #1: [F] MAC Address

The MAC Address configuration option controls the MAC address the adapter uses at power-on. The operating system device driver may change the MAC address of the BCM5880X SNIC controller at run-time (also known as a locally administered address or LAA), but this LAA will not modify the MAC address stored in NVRAM.

#### #108: [S] Max number of PF MSIX Vectors

This option sets the number of MSIX vectors that the device requests from the host on driver initialization for each PF.

#### #114: [S] Min Number of PF MSIX Vectors

This option internally set the number of minimal MSIX vectors required for the device initialization. Value 0 indicates a default value is selected. At the time of this writing, the default is 50.

#### #152: [P] Magic Packet WoL

If enabled, the controller asserts a wake pin upon reception of a WoL package with AMD WoL magic packet pattern.

#### #155: [P] DCBX Mode

The BCM5880X supports the Data Center Bridging eXchange protocol via IEEE mode or CEE mode.

#### #157: [P] MF Mode

This option is used to place the device into different modes that allow the device to appear as a single-function device or a multifunction device. [Table 2](#) lists the modes that are valid for this option. The default value is 0 (MF allowed).

**Table 2: Force SF Mode (Option #73)**

| Value | Mode       | Description                                                   |
|-------|------------|---------------------------------------------------------------|
| 0     | MF Allowed | Single-function device until activated by an external switch. |
| 1     | Forced SF  | The device is a single-function device.                       |
| 2     | NPAR1.0    | NPAR v1.0 mode.                                               |
| 3     | NPAR1.5    | NPAR v1.5 mode.                                               |
| 4     | NPAR2.0    | NPAR v2.0 mode.                                               |
| 5     | BD         | OEM specific NIC partition mode.                              |
| 6     | UFP        | OEM specific NIC partition mode.                              |
| 7     | AFEX       | OEM specific NIC partition mode.                              |

#### #166: [S] Enable PME Capability

This option is used to support S5 WOL in a Windows environment. This is necessary since Windows does not officially support S5 WOL and clears the PME Enable bit in the device for a S5 shutdown case.

#### #203: [P] Drv Link Speed



The Drv Link Speed configuration option controls the default link speed that is set when the host operating system driver (that is, Windows, Linux, and so on) is loaded, and when there is no user input.

#### **#205: [P] Firmware Link Speed**

The Firmware Link Speed configuration option controls the default link speed that is set in a pre-OS environment.

#### **#208: [P] EEE Power Saving Mode**

This configuration option controls the Energy Efficient Ethernet policy.

#### **#210: [P] Firmware Link Speed (for D3)**

This configuration sets the default link speed for the device while in D3 hot state (while the system is in a sleep state).

#### **#213: [P] Media Auto Detect**

This option enables a software feature to enhance SerDes link detection. This option must be set to enable for a proper link speed setting of optical transceivers if the link speed is set to auto-negotiation.

#### **#269: [P] LLDP Nearest Bridge**

This option enables/disables LLDP nearest bridge support. This is necessary for DCBX support.

#### **#270: [P] LLDP Nearest non-TPMR Bridge**

This option enables/disables LLDP nearest non-TPMR bridge support. This is necessary for DCBX support on such network.

#### **#312: [P] AN Mode**

This Auto-negotiation mode configuration allows selection between different Broadcom Multirate Backplane Ethernet mode (also known as Broadcom Auto-negotiation Mode [BAM]) on KR, and SGMII mode.

#### **#328: [P] Forward Error Correction**

Forward data error correction is a hardware SerDes feature. This option is useful when longer fiber cables are utilized. This option can be enabled or disabled.

#### **#329: [S] PHY selection {First PHY(0), Second PHY(1)}**

This option selects which of the two connected PHYs are used. This option is needed in a dual-media layout.

#### **#351: [F] MBA**

The Broadcom Multiple Boot Agent (MBA) is a legacy PC option ROM that allows the BCM5880X SNIC controller to be used for boot services such as PXE or iSCSI boot. The MBA firmware is generally programmed as part of BCM5880X SNIC controller NVRAM for NIC implementations and as part of the system BIOS flash for LOM implementations. This item enables or disables MBA.

#### **#352: [F] MBA Boot Type**

The MBA Boot Type configuration option controls what mechanism the MBA uses to insert itself into the list of bootable devices recognized by the system BIOS. The Auto selection attempts to identify whether the system BIOS supports the BIOS Boot Specification (or BBS). If that fails, the MBA tries to add itself as a boot device by hooking Interrupt 19h, followed by hooking Interrupt 18h. All other values will only attempt to boot through the selected method. Modify the value of this configuration item by entering the MBA setup menu and changing the MBA Boot Type there.

**#353: [F] MBA Delay Time (0–15)**

The MBA Delay Time configuration option controls the duration in seconds that the MBA banner displays during option ROM initialization. A value of 0 means that the MBA banner does not display.

**#354: [F] MBA Setup Hot Key**

The MBA Setup Hot Key configuration option controls what hot key is used to enter the MBA setup screen during option ROM initialization, <CTRL><S> or <CTRL><B>.

**#355: [F] MBA Hide Setup Prompt**

The MBA Hide Setup Prompt configuration option controls whether the MBA initialization banner prompts the end user to press the setup key to enter the MBA setup screen.

**#357: [F] MBA VLAN Value**

The MBA VLAN Value configuration option controls the value of the VLAN tag used by the BCM5880X SNIC controller when executing the MBA host software (such as the PXE or iSCSI boot ROM). Modify the value of this configuration item by entering the MBA setup menu and changing the value of the VLAN tag there. This is a 16-bit field and includes both the VLAN ID as well as the QOS bits.

**#358: [F] MBA VLAN**

The MBA VLAN configuration option controls whether a VLAN tag will be used by the BCM5880X SNIC controller when executing the MBA host software (such as the PXE or iSCSI boot ROM). Modify the value of this configuration item by entering the MBA setup menu and enabling/disabling a VLAN tag there.

**NOTE:** #357 and #358 only apply to PXE boot in legacy BIOS mode. These options currently do not apply in uEFI BIOS mode.

**#359: [F] MBA Link Speed**

The MBA Link Speed configuration option controls the link speed negotiated by the BCM5880X SNIC controller when executing the MBA host software (such as the PXE or iSCSI boot ROM). Modify the value of this configuration item by entering the MBA setup menu and changing the link speed there. [Table 3](#) lists the supported values.

**Table 3: Supported MBA Link Speed Values**

| Value | Link Speed       |
|-------|------------------|
| 0     | Auto-negotiation |
| 1     | 1G               |
| 2     | 10G              |
| 3     | 25G              |
| 4     | 40G              |
| 5     | 50G              |
| 6     | 100G             |

**#360: [F] MBA Boot Retry Count**

The MBA can retry booting if it fails for any reason. This option can be set to configure the Multiboot Agent (MBA) or PXE boot feature of the BCM5880X SNIC controller to retry a boot multiple times.

**#361: [F] MBA Boot Protocol**

The MBA Boot Protocol configuration option controls the boot protocol used by the MBA. Modify the value of this configuration item by entering the MBA setup menu and changing the boot protocol option there. BCM5880X SNIC controller supports PXE and iSCSI boot protocol.

**#401: [S] Enable SRIOV**

This option enables the SR-IOV characteristic of the device.

**#404: [F] Number of VFs per PF**

The number of VFs per PFs is an integer value that configures the VF/PF relationship of the device when in SR-IOV mode. Currently the number of VFs per PF can be set to up to 128 VFs when the device is configured with a single PF, or up to eight VFs per PF when a maximum of 16 PFs are configured. The default value is 16 VFs per PF.

**#406: [F] MSI-X Vectors per VF**

This configuration sets the number of MSI-X vectors allocated per VF. The default is 16. Each PF can be assigned up to 128 MSIX vectors.

**#457: [P] Secondary EXT PHY Link Speed**

The value of this option is the default link speed for the second external PHY.

**#459: [P] Secondary EXT Firmware WoL Link Speed**

This configurable item is the default value of the link speed configuration for the management firmware and for WoL of the second external PHY.

**#501: [F] NIC Partition Func BW Weight**

This is the relative weight of this NPAR compared to the other NPARs of a port. The values are 1–100. If one NPAR is set to 1 and another is set to 2, the NPAR with the value of 2 is weighted twice as much as the one with a value of 1. Therefore, it gets twice the bandwidth allocated to the first NPAR.

**#502: [F] NIC Partition Func BW Max**

This item is the maximum bandwidth allowed for this NPAR. The value is a percentage of the total bandwidth available on the port.

**#503: [P] Number of Partitions Per Port**

This item configures the number of NIC partitions per port.

**#506: [F] Support RDMA**

This configurable indicates if this NPAR PF supports RDMA. Current implementation allows up to 2 RDMA-enabled NPAR PFs per physical port.

**#508: [F] BW Reservation Valid**

This configuration indicates if option 507 is honored by the firmware.

**#509: [F] BW Limit Valid**

This configuration indicates if the bandwidth limit in option 507 is honored by firmware.

**#751: [U] Host to OOBM**

This configuration enables or disables Host to OOMB.

**#752: [U] OOBM Telnet Port**

This configures the TCP port to use for remote management.

**#753: [U] OOBM Telnet Timeout**

This configuration specifies in seconds the idle timeout for the remote management session. When a remote management session remains idle for this amount of time the session is terminated.

**#754: [U] OOBM AP UART Enable**

This configuration enables or disables AP(MAIA) UART.

**#757: [U] OOBM Syslog Port**

This configuration specifies the TCP port to use for the Syslog server.

**#758: [U] OOBM Syslog Server IPv4 Address**

This configuration specifies the IPv6 address to be used for the Syslog server interface.

**#761: [U] OOBM Debug**

This configuration enables or disables Debug.

**#762: [U] OOBM MAC Address Port 0**

This configuration specifies the MAC address to be used for the remote management interface.

**#763: [U] OOBM IPv4 Address Port 0**

This configuration specifies the IPv4 address to be used for the remote management interface.

**#764: [U] OOBM IPv4 Subnet Mask Port 0**

This configuration specifies the subnet mask to be used for the remote management interface.

**#765: [U] OOBM IPv4 Default Gateway Port 0**

This configuration specifies the IPv4 default gateway address to be used for the remote management interface.

## Appendix A: Acronyms and Abbreviations

Table 4 lists the acronyms and abbreviations used in this document.

For a more complete list of acronyms and other terms used in Broadcom documents, go to:

<http://www.broadcom.com/press/glossary.php>.

**Table 4: Acronyms and Abbreviations**

| Term    | Description                              |
|---------|------------------------------------------|
| BAM     | Broadcom Auto-negotiation Mode           |
| BBS     | BIOS Boot Specification                  |
| bnxtnvm | Broadcom diagnostic utility              |
| CLP     | Command Line Protocol                    |
| HWRM    | Hardware Resource Manager                |
| LLDP    | Link Layer Discovery Protocol            |
| MBA     | Multiple Boot Agent                      |
| NVRAM   | Non-volatile RAM                         |
| PMCSR   | Power Management Control/Status Register |
| RDMA    | Remote Direct Memory Access              |
| RoCE    | RDMA over Converged Ethernet             |
| VF      | Virtual Function                         |

## Revision History

### **5880X-PS225-AN302; January 31, 2020**

**Updated:**

- [Typical NVRAM Operations](#)

### **5880X-PS225-AN301; March 13, 2019**

**Updated:**

- Typical NVRAM Operations

### **5880X-PS225-AN300; December 11, 2018**

Initial Release

